

Vigenère Cipher Task Solutions

Task 1 Solution: “SECURITYALERT”

Keyword: “LOCK”

Step-by-step calculation:

Plaintext: S E C U R I T Y A L E R T
Key: L O C K L O C K L O C K L

$S(19) + L(12) = 31 - 26 = 5 \rightarrow E$
 $E(5) + O(15) = 20 \rightarrow T$
 $C(3) + C(3) = 6 \rightarrow F$
 $U(21) + K(11) = 32 - 26 = 6 \rightarrow F$
 $R(18) + L(12) = 30 - 26 = 4 \rightarrow D$
 $I(9) + O(15) = 24 \rightarrow X$
 $T(20) + C(3) = 23 \rightarrow W$
 $Y(25) + K(11) = 36 - 26 = 10 \rightarrow J$
 $A(1) + L(12) = 13 \rightarrow M$
 $L(12) + O(15) = 27 - 26 = 1 \rightarrow A$
 $E(5) + C(3) = 8 \rightarrow H$
 $R(18) + K(11) = 29 - 26 = 3 \rightarrow C$
 $T(20) + L(12) = 32 - 26 = 6 \rightarrow F$

Final answer: ETFFDXWJMAHCF

Task 2 Solution: “The Safe is Open”

Keyword: “VAULT”

Step-by-step calculation:

Plaintext: T h e S a f e i s O p e n
Key: V A U L T V A U L T V A U L
(spaces kept in original positions)

$T(20) + V(22) = 42 - 26 = 16 \rightarrow P$
 $h(8) + A(1) = 9 \rightarrow I$
 $e(5) + U(21) = 26 \rightarrow Z$
[space kept]
 $S(19) + L(12) = 31 - 26 = 5 \rightarrow E$
 $a(1) + T(20) = 21 \rightarrow U$
 $f(6) + V(22) = 28 - 26 = 2 \rightarrow B$
 $e(5) + A(1) = 6 \rightarrow F$
[space kept]
 $i(9) + U(21) = 30 - 26 = 4 \rightarrow D$
 $s(19) + L(12) = 31 - 26 = 5 \rightarrow E$
[space kept]
 $O(15) + T(20) = 35 - 26 = 9 \rightarrow I$
 $p(16) + V(22) = 38 - 26 = 12 \rightarrow L$
 $e(5) + A(1) = 6 \rightarrow F$
 $n(14) + U(21) = 35 - 26 = 9 \rightarrow I$

Final answer: Piz Eubf De Ilfi

Task 3 Solution: Decryption of “KPXVA QICKR ZTROI”

Keyword: “GOLD”

For decryption, we subtract the key values instead of adding:

Ciphertext: K P X V A Q I C K R Z T R O I
Key: G O L D G O L D G O L D G O L

$K(11) - G(7) = 4 \rightarrow E$
 $P(16) - O(15) = 1 \rightarrow A$
 $X(24) - L(12) = 12 \rightarrow S$
 $V(22) - D(4) = 18 \rightarrow T$
 $A(1) - G(7) = 20 \rightarrow E$
[space kept]
 $Q(17) - O(15) = 2 \rightarrow R$
 $I(9) - L(12) = 23 \rightarrow W$
 $C(3) - D(4) = 25 \rightarrow Y$
 $K(11) - G(7) = 4 \rightarrow E$
 $R(18) - O(15) = 3 \rightarrow T$
[space kept]
 $Z(26) - L(12) = 14 \rightarrow N$
 $T(20) - D(4) = 16 \rightarrow O$
 $R(18) - G(7) = 11 \rightarrow W$
 $O(15) - O(15) = 0 \rightarrow Z$
 $I(9) - L(12) = 23 \rightarrow W$

Decrypted message: EASTE RwyET NOWZW

Verification Process

For each solution: 1. Check that all mathematical calculations are correct 2. Verify wrap-around (subtraction of 26) is done correctly 3. Confirm key repetition pattern is maintained 4. Double-check that spaces and capitalization are preserved where required

Notes on Common Errors to Avoid:

1. Remember to subtract 26 when sum exceeds it
2. Keep track of key repetition carefully
3. For decryption, subtract key values instead of adding
4. Maintain original spacing and capitalization when required
5. Verify by re-encrypting the decrypted message